Production-Function Estimation,
Productivity, and Markups

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Abstract

This paper estimates a value-added Cobb–Douglas production function on a 1973–1988 panel of 2,971 firm-year observations across 121 manufacturing industries, then uses the estimates to recover firm productivity and markups under six estimators: pooled OLS, two-way fixed effects, and Olley–Pakes with and without a selection correction. Relative to OLS, Olley–Pakes lowers the capital elasticity (0.32 versus 0.43), the opposite direction from the original telecoms-equipment result, because multi-industry pooling and a flexible first-stage polynomial absorb cross-industry variation that OLS attributes to capital. The naive estimators get the productivity time trend backwards: OLS and fixed effects imply mean productivity rose over 1973–1988, while Olley–Pakes, conditioning on survival, recovers a mild decline and roughly 15% wider cross-firm dispersion. Estimator choice moves the level of inferred markups by up to a factor of 1.8 but leaves the industry ranking, time path, and dispersion unchanged. The exercise measures how much the selection correction actually changes substantive conclusions: it flips the productivity trend but not the markup structure.

Keywords: production-function estimation; Olley–Pakes; total factor productivity; markups; selection bias; firm heterogeneity.
JEL: D24, C23, L11, L60.

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Plain-Language Summary

Economists constantly estimate two things about firms: how productive they are, and how much they mark up prices over cost. Both rest on a "production function" linking inputs (labor, capital) to output. This paper estimates one six different ways on 2,971 U.S. manufacturing firms tracked from 1973 to 1988, to see how much the method matters.

The main lesson is about survivor bias. Firms that fail drop out of the data, and survivors tend to be the more productive ones. A naive method mistakes this weeding-out for genuine productivity growth. The Olley–Pakes method accounts for which firms are likely to exit, and once it does, the apparent "growth" reverses into a mild decline. The growth was an illusion created by the losers leaving the sample.

For markups, the method matters less. The estimated level of market power shifts with the method, but which industries are most marked up (drugs and autos at the top, computers at the bottom), how markups move over time, and how spread out they are all stay the same.

The honest caveat: the markup formula leans entirely on the labor elasticity, and the "computers" industry here is really two eras, mainframe IBM and PC clones, glued together, which pushes its measured markup implausibly below cost. The practical takeaway is that the selection correction earns its keep when it would change a conclusion. Here it flips the productivity story but not the markup story.

CONTENTS

OVERVIEW

The dataset prodfncData.dta records $N = 2{,}971$ firm-year observations on 1,400 distinct firms across 121 three-digit SIC industries, observed in four waves: 1973, 1978, 1983, and 1988 (encoded 73/78/83/88). For each firm-year the data report a firm identifier index, industry code sic3, year yr, log deflated value added (ldsal, the production-function output y), log employment (lemp, the labor input ℓ), log deflated capital (ldnpt, the state variable k), log deflated R&D stock (ldrst) and R&D flow (ldrnd), log deflated investment (ldinv, the proxy variable i used in Olley–Pakes), and a year-level wage index W_t. There are no missing values on any variable. Wave sizes are imbalanced (644, 890, 751, 686 for 1973–1988), which already signals that entry and exit, not just survey timing, drive the panel structure.

The document proceeds in three parts. **Part 1** characterizes the panel: industry distribution, firm-size heterogeneity in the modal industry, panel-balance decomposition, and full-sample-vs-balanced-sample summary statistics that establish the selection problem. **Part 2** estimates the production function: four naive OLS/FE specifications, the survival probit that powers the Olley–Pakes selection correction, and the two-stage OP estimator with and without selection control. **Part 3** uses the estimates for policy analysis: recovering firm-level productivity $\hat{\omega}_{it}^{m}$ and implied markups $\hat{\mu}_{it}^{m}$ for each of the six estimators, comparing distributions, time paths, industry heterogeneity, and cross-method robustness.

All computations were performed in R (version $\geq$ 4.3). Code is reproducible by sourcing Dalis_Abdallah_main.R, which runs the six numbered scripts 01–06 in order. See Appendix A for the file map.

Part 1: Descriptives

1 NUMBER OF INDUSTRIES AND THE MODAL INDUSTRY

There are **121** distinct 3-digit SIC industries in the panel. The industry with the most firm-year observations is **SIC 357**, *Computer and Office Equipment*, with $n = 195$ observations contributed by 107 distinct firms (Table 1). That is, roughly 6.6% of the panel sits inside SIC 357 alone. The top three industries (357, 382 Laboratory Instruments, 367 Electronic Components) account for 543 observations, or 18.3% of the sample, so the panel concentrates heavily in tech-intensive durable-goods manufacturing, which is exactly the set of industries where R&D-driven productivity heterogeneity, the object the Olley–Pakes estimator was built to recover, is most economically interesting.

2 FIRM-SIZE DISTRIBUTION IN THE MODAL INDUSTRY

Figure 1 plots the histogram of log deflated value added (ldsal) for the 195 firm-year observations in SIC 357. The distribution has a mean of 4.42, a standard deviation of 2.14, and ranges from -0.86 to 11.15, i.e. value-added per firm spans roughly $e^{12} \approx 162{,}000$ times across the

Table 1: Top 10 industries (3-digit SIC) by number of firm-year observations. Total observations = 2971.

SIC3	Obs.	Share (%)
357	195	6.56
382	189	6.36
367	159	5.35
283	137	4.61
371	122	4.11
356	117	3.94
366	113	3.80
384	93	3.13
372	75	2.52
291	74	2.49

industry. The bulk of mass sits between `ldsal` $\in [2, 7]$, with a clear right tail that thins out steadily toward the maximum. The shape is roughly bell-shaped on the log scale, consistent with a heavy-tailed firm-size distribution on the raw scale, in line with the Gibrat/Zipf stylized facts. Practically, this matters for production-function estimation: the wide support in y comes with a correspondingly wide support in k and ℓ, which is what identifies the input elasticities in OLS and disciplines the polynomial inversion in Olley–Pakes. A degenerate (narrow) industry would have collinear inputs and almost no power to separate β_ℓ from β_k.

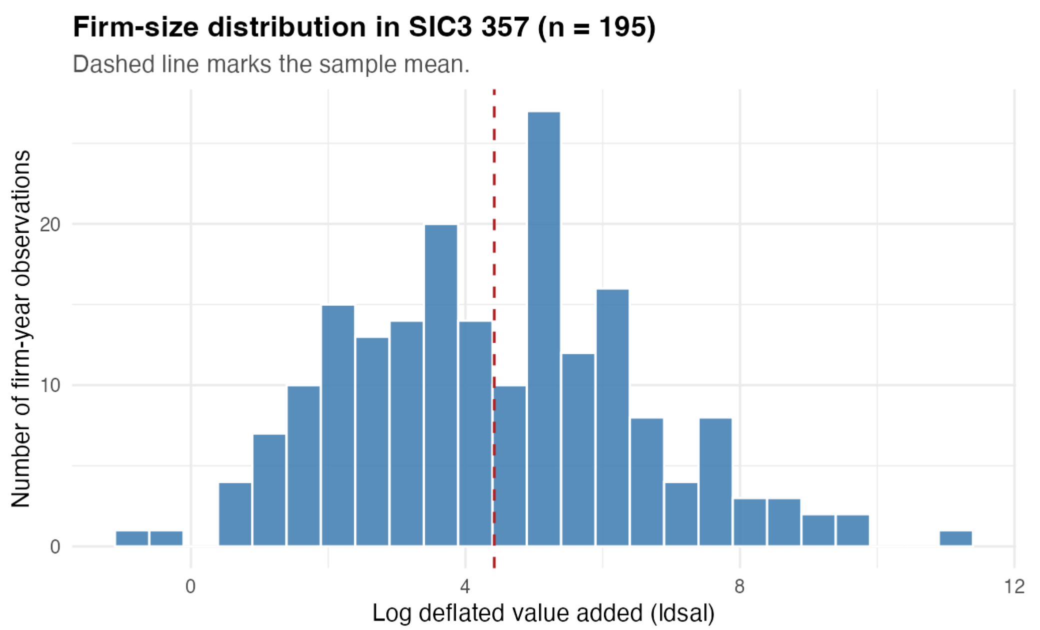

Figure 1: Distribution of log deflated value added (`ldsal`) across firm-year observations in SIC 357 (Computer and Office Equipment). $n = 195$ observations from 107 distinct firms. Dashed line marks the sample mean (4.42).

3 PANEL BALANCE

The panel is unbalanced. If it were balanced, every one of the 1,400 firms would appear in all four waves, yielding $1{,}400 \times 4 = 5{,}600$ observations, but we observe only 2,971. Table 2 decomposes the unbalance by number of waves per firm. Only 214 firms (15.3% of firms; 28.8% of observations) appear in all four waves; 531 firms (37.9%) appear in exactly one wave, 381 in two, and 274 in three.

Table 2: Distribution of the number of years each firm is observed. The panel runs over four waves (1973, 1978, 1983, 1988). Firms observed in all four waves form the balanced panel.

Years observed	Firms	Firms (%)	Firm-yr obs.	Obs. (%)
1	531	37.93	531	17.87
2	381	27.21	762	25.65
3	274	19.57	822	27.67
4	214	15.29	856	28.81

This is not random missingness. The four-wave coverage corresponds to firms that survived the entire 1973–1988 window, through the 1973–75 stagflation, the Volcker disinflation of the early 1980s, and the mid-1980s industrial restructuring, while one- and two-wave firms include both new entrants and firms that exited before 1988. This pattern is the exact reason Olley and Pakes (1996) build a survival/selection correction into the estimator: pooling OLS on the unbalanced sample mixes survivors with exiters whose unobserved productivity was correlated with the inputs they chose, producing the canonical OLS upward bias on capital. The selection problem is therefore not a technicality of this dataset; it is the central econometric problem the rest of the problem set is designed to address.

4 SUMMARY STATISTICS, FULL SAMPLE

Table 3 reports the count, mean, standard deviation, and key quantiles for each variable across the full unbalanced panel ($N = 2{,}971$). All input and output variables are reported in logs of deflated levels, which compresses the right tail but preserves the wide dispersion characteristic of firm-level manufacturing data: `ldsal` ranges from -0.86 to 11.70, and `ldnpt` from -1.39 to 11.11. Mean log value added (5.67) exceeds mean log capital (4.47) which in turn exceeds mean log employment (1.26), consistent with capital being measured in deflated dollar units and employment in raw headcount, not a structural fact about returns to scale. R&D stock means ($\overline{\texttt{ldrst}} = 3.40$) exceed R&D flow means ($\overline{\texttt{ldrnd}} = 1.79$) by roughly the magnitude one would expect from a perpetual-inventory cumulation. Investment (`ldinv`, mean 2.67) is the proxy variable that Olley–Pakes will invert to recover the unobserved productivity shock, and its full support $[-3.84,\ 8.99]$ is what gives the inversion power. Wages average \$23.17/hour with low variation (SD = 3.61), reflecting the fact that the wage series is essentially year-only.

5 SUMMARY STATISTICS, BALANCED PANEL, AND SELECTION BIAS

Table 4 reports the same statistics restricted to the 856 observations from the 214 four-wave survivors. **The balanced panel is systematically different from the full sample, and the**

Table 3: Summary statistics, full unbalanced panel ($N = 2971$ firm-year observations, 1400 firms across 1973–1988).

Variable	N	Mean	SD	Min	P25	Median	P75	Max
y log deflated value added (ldsal)	2971	5.673	1.961	-0.857	4.251	5.529	7.084	11.698
ℓ log employment (lemp)	2971	1.259	1.775	-3.772	-0.025	1.114	2.632	6.732
k log deflated capital (ldnpt)	2971	4.469	2.217	-1.389	2.946	4.212	6.001	11.112
log deflated R&D capital (ldrst)	2971	3.401	2.029	-4.287	1.990	3.177	4.792	9.966
log deflated R&D (ldrnd)	2971	1.788	2.052	-5.313	0.354	1.631	3.189	8.432
i log deflated investment (ldinv)	2971	2.675	2.170	-3.844	1.124	2.506	4.196	8.989
Wage	2971	23.175	3.614	18.375	21.875	21.875	23.625	28.875

differences point in exactly the direction Olley–Pakes warns about.

Table 4: Summary statistics, balanced panel ($N = 856$ firm-year observations, 214 firms observed in all four waves).

Variable	N	Mean	SD	Min	P25	Median	P75	Max
y log deflated value added (ldsal)	856	6.914	1.838	1.664	5.564	7.057	8.267	11.698
ℓ log employment (lemp)	856	2.413	1.622	-2.071	1.133	2.652	3.627	6.732
k log deflated capital (ldnpt)	856	5.916	2.057	0.806	4.276	6.032	7.422	11.112
log deflated R&D capital (ldrst)	856	4.886	1.930	0.062	3.292	5.109	6.286	9.966
log deflated R&D (ldrnd)	856	3.222	1.994	-2.711	1.642	3.403	4.673	8.432
i log deflated investment (ldinv)	856	4.069	2.055	-2.076	2.466	4.234	5.568	8.888
Wage	856	23.188	3.791	18.375	21.000	22.750	24.938	28.875

Mean differences (balanced minus full sample). Across every input and output variable except wages, balanced-panel means are substantially *larger* than full-sample means: $\Delta\overline{\texttt{ldsal}} = +1.24$, $\Delta\overline{\texttt{lemp}} = +1.15$, $\Delta\overline{\texttt{ldnpt}} = +1.45$, $\Delta\overline{\texttt{ldrst}} = +1.48$, $\Delta\overline{\texttt{ldrnd}} = +1.43$, $\Delta\overline{\texttt{ldinv}} = +1.39$. A shift of roughly $+1.4$ on the log scale corresponds to firms that are about $4\times$ larger in every dimension. Wages move by only $+0.01$, which is expected because wages are essentially time-only and the balanced panel's year mix is mechanically uniform.

Dispersion shrinks. Standard deviations are uniformly smaller in the balanced panel (ΔSD $\texttt{ldsal} = -0.12$, ΔSD $\texttt{lemp} = -0.15$, ΔSD $\texttt{ldnpt} = -0.16$, etc.). Conditioning on survival drops both the lower tail (small firms more likely to exit) and the extreme upper tail (very large firms more likely to be acquired or restructured out of the sample), leaving a tighter cluster of mid-to-large incumbents.

Why this matters for estimation. The balanced panel is not a representative subsample of the underlying population; it is a selected sample of long-lived incumbents whose productivity draws were high enough to keep them above the exit threshold for fifteen consecutive years. Running OLS on the balanced sample throws away the variation in entry/exit that the data actually contain, while running OLS on the full unbalanced sample fails to correct for the fact that capital stock is correlated with the survival decision. Either approach produces biased input elasticities, but in different directions: balanced-OLS understates the role of capital because all the low-k exiters have been removed, while unbalanced-OLS understates it differently because exiters' low realized productivity is mechanically attributed to their (low) capital. The Olley–Pakes estimator in Part 2 will use the full unbalanced sample precisely so that the survival probability can be modeled directly via the first-stage probit on i and k, and so that the second-stage productivity recovery is conditioned on the firm having not yet exited. The descriptive contrast above is what tells us that the selection correction is not cosmetic, the survivors are systematically different along every input dimension.

Part 2: Estimation

Part 2 estimates the value-added Cobb–Douglas production function

$$y_{it} = \beta_\ell \ell_{it} + \beta_k k_{it} + \beta_a a_{it} + \omega_{it} + \eta_{it} \tag{1}$$

where ω_{it} is the unobserved Hicks-neutral productivity shock and η_{it} is an i.i.d. measurement/optimization error. Firm age a_{it} is constructed as a panel-tenure proxy equal to the firm's survey-wave sequence number, the running count of waves in which the firm has been observed, $a_{it} \in \{1, 2, 3, 4\}$ (waves are five years apart). True firm birth dates are not observed, so a is left-censored for any firm already operating at its first observed wave; this is the standard construction (Olley and Pakes, 1996) when birth is unobserved.

6 NAIVE OLS ESTIMATES

I estimate four naive specifications of the production function with ω omitted entirely from the regression equation:

$$ldsal_{it} = \beta_0 + \beta_\ell \, lemp_{it} + \beta_k \, ldnpt_{it} + \varepsilon_{it}, \tag{2}$$

with $\varepsilon_{it} = \omega_{it} + \eta_{it}$ collapsing the productivity term into the error. Two cuts are run on each panel: pooled OLS (treating ε_{it} as conditionally mean-zero) and OLS with industry (α_{sic3}) and year (γ_{yr}) fixed effects. Each is fit on both the unbalanced sample ($N = 2{,}971$) and the balanced sub-panel of four-wave survivors ($N = 856$). Standard errors are clustered at the firm (`index`) level in all four columns. Following the PS prompt, the naive specifications include only labor and capital; firm age is added in the OP estimation in Section 8 to match Olley and Pakes (1996) and Barrozo's lecture spec. The four-specification comparison reflects both this addition of age *and* the OP correction.

Coefficient direction across columns. Pooled OLS on the unbalanced panel yields $\hat{\beta}_\ell = 0.543$, $\hat{\beta}_k = 0.427$, with implied returns to scale (RTS) of 0.97. Adding industry and year fixed effects raises $\hat{\beta}_\ell$ to 0.771 and shrinks $\hat{\beta}_k$ to 0.189 in the unbalanced sample, with virtually unchanged RTS (0.96). The same direction, $\hat{\beta}_\ell$ up, $\hat{\beta}_k$ down, appears in the balanced sample under FE, just with attenuated magnitudes. The balanced-vs-unbalanced shift in the OLS-only columns (col. 1 $\rightarrow$ col. 2) moves $\hat{\beta}_k$ from 0.427 to 0.533 and $\hat{\beta}_\ell$ from 0.543 to 0.429: restricting to survivors raises the capital coefficient and lowers the labor coefficient, foreshadowing exactly the selection problem the OP estimator is designed to address.

The FE columns are not addressing the OP problem. Industry FE sweep out cross-industry differences in mean productivity; year FE absorb macro shocks. Neither touches within-firm simultaneity between input choices and ω_{it}, nor selection on ω that drives exit. So the OLS-to-FE

Table 5: Naive Cobb–Douglas production-function estimates

	(1) OLS unbalanced	(2) OLS balanced	(3) FE unbalanced	(4) FE balanced
Constant	3.0798***	2.7235***		
	(0.0610)	(0.1250)		
β_ℓ (Labor)	0.5433***	0.4289***	0.7714***	0.7065***
	(0.0242)	(0.0484)	(0.0239)	(0.0411)
β_k (Capital)	0.4272***	0.5335***	0.1891***	0.2713***
	(0.0190)	(0.0356)	(0.0215)	(0.0412)
Num.Obs.	2971	856	2961	856
R2	0.919	0.913	0.951	0.961
R2 Adj.	0.919	0.913	0.949	0.958
R2 Within			0.921	0.924
R2 Within Adj.			0.921	0.923

Dependent variable: log deflated value added (ldsal).
FE columns include 3-digit SIC industry and year fixed effects.

shifts here reflect omitted heterogeneity, not the OP biases: the jump in $\hat{\beta}_\ell$ from 0.54 to 0.77 is industry-level labor intensity being absorbed, and the drop in $\hat{\beta}_k$ to 0.19 is cross-industry capital-intensity dispersion (e.g., SIC 283 pharmaceuticals vs. SIC 371 motor vehicles) being soaked up by the industry intercepts. The OP estimator in Section 8 is what addresses the actual biases.

Ten observations drop out of column 3 as fixed-effect singletons (industries appearing only once); this is `fixest`'s default and doesn't affect the comparison.

7 SURVIVAL PROBIT

The Olley–Pakes selection correction requires a model of $P_{it} \equiv \Pr(\chi_{i,t+1} = 1 \mid k_{i,t+1}, a_{i,t+1}, J_t)$: the conditional probability that firm i survives into the next wave. I construct the survival indicator $survival_{it} = \mathbf{1}\{yr_{i,t+1} = yr_{it} + 5\}$, equal to one if the firm is observed in the very next five-year wave. The 686 observations in the 1988 wave have no $t+1$ inside the panel and are dropped, leaving $N = 2{,}285$ for the probit. The empirical survival rate is 0.657, with a clear declining trend across waves (0.761 for 1973 $\to$ 1978, 0.642 for 1978 $\to$ 1983, 0.587 for 1983 $\to$ 1988) consistent with the 1980s manufacturing shakeout.

Two probits are estimated. Spec 1 (Table 6) is a simple linear-in-(i,k) probit reported for coefficient interpretation:

$$\Pr(survival_{it} = 1) \;=\; \Phi\big(\delta_0 + \delta_i\, ldinv_{it} + \delta_k\, ldnpt_{it} + \alpha_{sic3} + \gamma_{yr}\big). \tag{3}$$

Spec 2 is a 4th-degree polynomial in $(ldnpt, ldinv, a)$ with the same fixed effects, used purely as a prediction model inside the OP second stage in Section 8. Its coefficients are not interpretable individually and are not reported; $\widehat{P}_{it}$ from Spec 2 is what enters OP2.

Table 6: Survival probit estimates

	Survival probit (Spec 1)
Constant	-0.0300
	(0.3051)
δ_i (log investment)	0.1524***
	(0.0464)
δ_k (log capital)	0.0359
	(0.0479)
Year = 1978	-0.3539***
	(0.0751)
Year = 1983	-0.5846***
	(0.0795)
Num.Obs.	2285
AIC	2796.1
BIC	3501.3
Log.Lik.	-1275.026

Dependent variable: $survival_{it} = 1$ if firm observed in wave $t + 5$.
the Olley–Pakes second stage in script 04; coefficients not reported.

Sign on investment, $\hat{\delta}_i = 0.152$ ($p = 0.001$). The investment coefficient is significantly positive, in line with the optimal-investment-monotonicity result (Pakes, 1994): investment is a monotone-increasing function of the unobserved productivity shock ω_{it} in the firm's dynamic program, so firms with higher current investment have, in expectation, higher current ω, higher continuation value, and therefore higher one-period-ahead survival probability. This is precisely the moment that makes investment a valid proxy for ω in the OP inversion.

Sign on capital, $\hat{\delta}_k = 0.036$ ($p = 0.45$). The capital coefficient is positive but statistically indistinguishable from zero. This is consistent with capital being only weakly informative about survival once investment is controlled for: investment captures the *current* productivity signal that determines the continuation decision, while capital is a sluggish state variable accumulated from prior periods. A non-finding on δ_k is a substantive result, not a failure: it means that in this dataset the selection correction inside OP will rely primarily on the variation in $\widehat{P}_{it}$ that comes from investment, not from capital. By implication, we should expect the OP1 (no selection) and OP2 (with selection) $\hat{\beta}_k$ estimates to differ less than they would in a dataset where capital strongly predicts exit.

Why this matters for the production function. The probit gives us $\widehat{P}_{it}$, which the OP second stage uses as a control function to break the correlation between $k_{i,t+1}$ and unobserved ω conditional on continuation. Without it, OLS attributes survivor-conditional productivity variation to the inputs and biases $\hat{\beta}_k$; with $\widehat{P}_{it}$ and the Markov assumption on ω, that channel is shut.

8 OLLEY–PAKES ESTIMATION

The OP estimator runs in two stages. The first stage uses the inversion result (Olley and Pakes, 1996) that, under monotonicity of investment in productivity, $\omega_{it} = h(k_{it}, i_{it}, a_{it})$ for some function h. Substituting into equation (1):

$$y_{it} \;=\; \beta_\ell\, \ell_{it} \;+\; \varphi(k_{it}, i_{it}, a_{it}) \;+\; \eta_{it}, \tag{4}$$

where $\varphi(\cdot) = \beta_k\, k + \beta_a\, a + h(i, k, a)$ collects the input contributions plus the inverted productivity term. The first stage approximates $\varphi(\cdot)$ by a 4th-degree polynomial in (k, i, a) with all interactions (`poly(ldnpt, ldinv, age, degree = 4, raw = TRUE)`) and recovers $\hat{\beta}_\ell$ as the OLS coefficient on labor. From this I back out $\hat{\varphi}_{it} = y_{it} - \hat{\beta}_\ell\, \ell_{it}$, which equals the polynomial fit plus the OLS residual and therefore inherits the appropriate moment structure.

The second stage uses the productivity Markov assumption to identify (β_a, β_k). Let $h_{it} \equiv \hat{\varphi}_{it} - \beta_a\, a_{it} - \beta_k\, k_{it}$; at the true parameters, $h_{it} = \omega_{it}$. Under $\omega_{i,t+1} = g(\omega_{it}, P_{it}) + \xi_{i,t+1}$ with ξ mean-uncorrelated with anything dated $\leq t$, the estimating moment is:

$$\underbrace{y_{i,t+1} - \hat{\beta}_\ell\, \ell_{i,t+1}}_{=\hat{\varphi}_{i,t+1}} - \beta_a\, a_{i,t+1} \;-\; \beta_k\, k_{i,t+1} \;=\; g\big(h_{it},\, [\,\widehat{P}_{it}\,]\big) \;+\; \xi_{i,t+1} \;+\; \eta_{i,t+1}. \tag{5}$$

The bracketed $\widehat{P}_{it}$ is in for OP2 and out for OP1. $g(\cdot)$ is a 4th-degree polynomial. For each candidate (β_a, β_k), the right-hand-side regression is fit by OLS, the SSR is extracted, and `optim` (Nelder–Mead, starting values $(0.1, 0.5)$) minimizes the SSR over (β_a, β_k). Standard errors are firm-cluster bootstrapped ($B = 200$ resamples, seed $= 514$), reapplying the full two-stage procedure each rep. Stage 1 has $N = 2{,}971$; stage 2 has $N = 1{,}502$ matched firm-year pairs (firms observed in two consecutive waves). First and second stages omit FE (the original OP paper includes year dummies in stage 1, a defensible robustness check not run here).

Table 7: Olley–Pakes production-function estimates. Cluster-bootstrap standard errors in parentheses (200 firm-resamples, seed = 514).

Coefficient	(1) OP1 (no selection)	(2) OP2 (with selection)
β_ℓ (Labor)	0.5322	0.5322
	(0.0244)	(0.0250)
β_a (Age)	0.3324	0.3755
	(0.0340)	(0.0409)
β_k (Capital)	0.3604	0.3199
	(0.0260)	(0.0267)
Stage-2 observations	1502	1502
Optim convergence	yes	yes

Table 8 consolidates the key coefficient $\hat{\beta}_k$ across all six specifications for ease of comparison;

this six-row summary is the reference table for everything that follows in Part 3.

Table 8: $\hat{\beta}_k$ across all six specifications (PS2). Bootstrap or clustered standard errors in parentheses where available.

Estimator	$\hat{\beta}_\ell$	$\hat{\beta}_k$	$\hat{\beta}_a$
(1) OLS unbalanced	0.543 (0.024)	0.427 (0.019)	–
(2) OLS balanced	0.429 (0.048)	0.533 (0.036)	–
(3) FE unbalanced	0.771 (0.024)	0.189 (0.021)	–
(4) FE balanced	0.707 (0.041)	0.271 (0.041)	–
(5) OP1 (no selection)	0.532 (0.024)	0.360 (0.026)	0.332 (0.034)
(6) OP2 (with selection)	0.532 (0.025)	0.320 (0.027)	0.376 (0.041)

Labor coefficient. $\hat{\beta}_\ell$ moves from 0.543 (pooled OLS) to 0.532 (OP), a downward shift of 0.011. The Marschak and Andrews (1944) simultaneity logic predicts that OLS overstates $\hat{\beta}_\ell$ because labor choice ℓ_{it} is correlated with the contemporaneous productivity shock ω_{it} (high-ω firms hire more). The OP first stage absorbs that correlation by inverting ω out of the regression via the investment proxy, so OP $\hat{\beta}_\ell$ should be lower than OLS. The direction is correct, but the magnitude is tiny, only one standard error of bootstrap noise. The substantive implication is that labor simultaneity is essentially absent at this dataset's five-year wave frequency: employment is sufficiently persistent (or pre-committed) that the within-wave correlation between ℓ_{it} and the wave-t productivity innovation is small. This is consistent with the literature finding that the Marschak–Andrews bias matters most in high-frequency (monthly or quarterly) panels where labor can be flexibly adjusted to the current shock.

Capital coefficient. $\hat{\beta}_k$ moves from 0.427 (OLS unbalanced) to 0.360 (OP1) to 0.320 (OP2), both OP estimates below OLS. **This is the opposite direction from Olley and Pakes (1996) themselves**, whose telecoms-equipment results gave OP $\hat{\beta}_k = 0.328$ vs. OLS 0.219. The capital-bias direction isn't theoretically pinned down, it depends on the joint distribution of (k, ω, ℓ). Two features of this dataset likely matter: (i) the panel pools 121 SIC3 industries vs. OP's single industry, so the first-stage polynomial in (k, i, a) is absorbing cross-industry heterogeneity that OLS attributes to k; (ii) the 34-monomial first stage is flexible enough to soak up variation that a linear-in-k OLS leaves for the capital coefficient.

OP1 $\rightarrow$ OP2 lowers $\hat{\beta}_k$ by 0.040, the predicted direction: among survivors, high-capital firms continued for reasons correlated with productivity, so OP1 over-attributes that to k, and adding $\widehat{P}_{it}$ in $g(\cdot)$ strips it out. The magnitude is small, consistent with Q7: capital was only weakly predictive of survival once investment was controlled for, so the selection correction doesn't bite hard here. I don't claim the OP1-vs-OP2 difference is statistically significant: the bootstraps were run independently, so $\mathrm{SE}(\hat{\beta}_k^{\mathrm{OP2}} - \hat{\beta}_k^{\mathrm{OP1}})$ isn't directly recoverable, and the two 95% CIs overlap heavily (OP1 $[0.310, 0.410]$, OP2 $[0.269, 0.367]$).

Age coefficient. $\hat{\beta}_a = 0.332$ (OP1) and 0.376 (OP2), with bootstrap standard errors of 0.034 and 0.041, precisely estimated and bounded well away from zero. Age is measured in survey *waves* (each five years apart), so the coefficient says each additional observed wave of panel tenure is associated with roughly 33–38% more log value added, controlling for inputs, equivalently about 7% per year, the same economic magnitude regardless of whether age is scaled in waves or in years. This is plausibly learning-by-doing or accumulated organizational capital. Because a counts observed waves rather than true firm age, it is a left-censored panel-tenure proxy: a firm already mature when it first enters the panel is recorded at wave 1 alongside a genuine entrant, so I identify *within-panel* tenure effects only, and the estimate should not be over-interpreted as a structural firm-age effect.

Returns to scale. $\hat{\beta}_\ell + \hat{\beta}_k = 0.97$ for OLS unbalanced, 0.96 for FE unbalanced, and 0.89 (OP1) / 0.85 (OP2). Returns to scale are summed over the scalable inputs, labor and capital, only; firm age a is a state variable rather than an input that scales with firm size, so $\hat{\beta}_a$ does not enter the RTS sum. The OP estimates indicate mild decreasing returns at the firm level, which fits the value-added manufacturing literature.

One flag on the polynomial probit. It has 155 coefficients on 2,285 obs, with AIC slightly above the simple probit (2,810 vs. 2,796). The polynomial is overfitting noise more than it's catching nonlinearity. That's fine for OP, we only use it as a prediction function for $\widehat{P}_{it}$, but the marginal informativeness over the simple probit is essentially zero here.

Part 3: Policy Analysis

Part 3 uses the six estimators from Part 2, two pooled OLS, two two-way fixed-effect (FE), and OP with (OP2) and without (OP1) the selection correction, to recover firm-level total factor productivity $\hat{\omega}_{it}$ and to compute implied markups $\hat{\mu}_{it}$ on the full unbalanced panel of $N = 2{,}971$ firm-year observations. For each estimator m I define

$$\hat{\omega}^m_{it} \;=\; y_{it} \;-\; \hat{\beta}^m_\ell\, \ell_{it} \;-\; \hat{\beta}^m_k\, k_{it} \;-\; \hat{\beta}^m_a\, a_{it}, \tag{6}$$

where $\hat{\beta}^m_a \equiv 0$ for the OLS/FE estimators (age enters only the OP specifications). Markups are computed by the cost-minimization production approach of Hall (1988) as operationalized by De Loecker and Warzynski (2012):

$$\hat{\mu}^m_{it} \;=\; \hat{\beta}^m_\ell \cdot \frac{P_{it}\, Q_{it}}{W_t\, L_{it}} \;=\; \hat{\beta}^m_\ell \cdot \frac{\exp(ldsal_{it})}{W_t \cdot \exp(lemp_{it})}, \tag{7}$$

under the assumption that labor is the within-period flexible input (capital is the state). The level of $\hat{\mu}$ inherits the unit normalization of the wage series; ratios, dispersions, time paths, and cross-method comparisons are unit-invariant, and I lean on those.

9 IMPLIED PRODUCTIVITY

Table 9 reports the cross-sectional distribution of $\hat{\omega}^m_{it}$ for each of the six estimators; Figure 2 shows the histograms; Table 10 and Figure 3 show the year-by-year means.

Table 9: Distribution of estimator-implied productivity $\hat{\omega}_{it}$ across the full unbalanced sample ($N = 2{,}971$). $\hat{\omega}^m_{it} = y_{it} - \hat{\beta}^m_\ell\, \ell_{it} - \hat{\beta}^m_k\, k_{it} - \hat{\beta}^m_a\, a_{it}$; for OLS/FE estimators $\hat{\beta}^m_a \equiv 0$.

Estimator	N	Mean	SD	P10	P50	P90	Min	Max
OLS unbalanced	2971	3.080	0.559	2.573	3.111	3.646	-0.422	5.859
OLS balanced	2971	2.749	0.565	2.214	2.780	3.325	-0.852	5.767
FE unbalanced	2971	3.857	0.596	3.306	3.897	4.452	0.330	6.037
FE balanced	2971	3.571	0.573	3.053	3.600	4.143	0.132	5.990
OP1 (no selection)	2971	2.782	0.611	2.118	2.817	3.454	-0.446	5.497
OP2 (with selection)	2971	2.883	0.640	2.166	2.912	3.594	-0.379	5.443

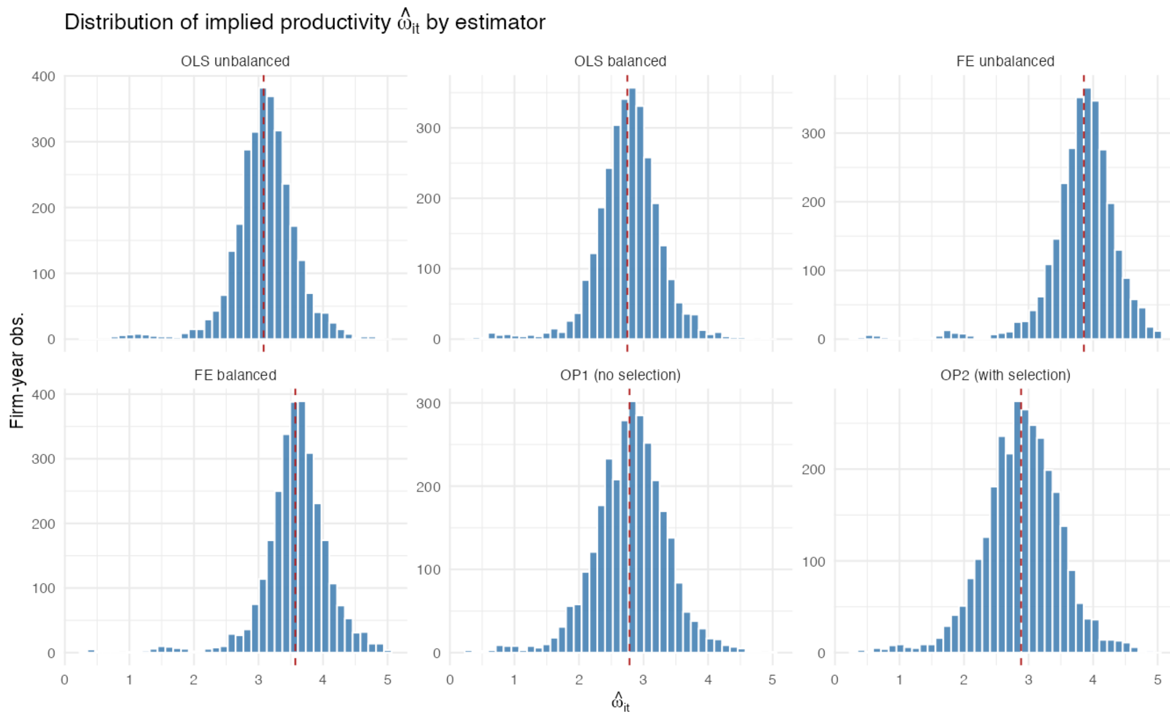

Figure 2: Six-panel histogram of recovered productivity $\hat{\omega}^m_{it}$, $N = 2{,}971$ firm-year observations. Vertical dashed line marks the estimator-specific sample mean. Bin range trimmed at the 0.5/99.5 percentiles.

Level differences across estimators are mechanical, not substantive. Mean $\hat{\omega}$ ranges from 2.75 (OLS balanced) to 3.86 (FE unbalanced). These differences reflect what the slope coefficients absorb: FE specifications place a smaller share of y on the inputs ($\hat{\beta}_\ell + \hat{\beta}_k = 0.96$ in OLS-unbal vs. 0.96 in FE-unbal but with the labor coefficient 0.77 instead of 0.54), so more of y is left in the residual. OLS-balanced anchors lowest because it loads the most weight on k ($\hat{\beta}_k = 0.53$) among the six specifications, and balanced-panel firms have higher mean k than the full sample (Section 5: $\Delta\overline{ldnpt} = +1.45$). The estimator-implied productivity *level* is not the object of interest; the cross-sectional dispersion and the time path are.

Dispersion: OP recovers more cross-firm heterogeneity than OLS or FE. The OP1 and OP2 distributions have IQRs of 0.68 and 0.73, while all four naive specifications cluster at IQR $\in [0.51, 0.55]$. Equivalently, $\text{SD}(\hat{\omega}^{\text{OP2}}) = 0.64$ vs. $\text{SD}(\hat{\omega}^{\text{OLS-unbal}}) = 0.56$, a $\approx 15\%$ wider spread. This pattern is the predicted one. Naive OLS biases $\hat{\beta}_k$ upward by attributing exiters' realized low ω to their low k; that channel inflates the apparent role of inputs in explaining y and *compresses* the residual $\hat{\omega}$. The OP procedure shuts that channel by inverting ω out of the regression via the investment proxy, so the residual recovers more of the true productivity heterogeneity. OP2's spread exceeds OP1's by a further 5% because the selection correction lowers $\hat{\beta}_k$ from 0.360 to 0.320, leaving an even larger residual for capital-rich firms (a known feature of selection-corrected productivity).

Productivity over time: the central policy finding. Table 10 and Figure 3 show the divergence sharply. The naive estimators (OLS, FE, both balanced and unbalanced) all imply that mean productivity *rose* from 1973 to 1988 by between $+0.35$ and $+0.39$ log points. The Olley–Pakes estimators imply the opposite: $\overline{\omega}^{\text{OP1}}$ *falls* from 3.00 in 1973 to 2.80 in 1988 (-0.20), and $\overline{\omega}^{\text{OP2}}$ falls from 3.16 to 2.86 (-0.30).

Table 10: Mean implied productivity $\overline{\omega}_t^m$ by year, all six estimators.

Estimator	1973	1978	1983	1988	Δ(88−73)
OLS unbalanced	2.993	3.015	2.976	3.359	+0.367
OLS balanced	2.672	2.699	2.626	3.021	+0.349
FE unbalanced	3.759	3.759	3.792	4.146	+0.387
FE balanced	3.471	3.485	3.494	3.861	+0.390
OP1 (no selection)	3.004	2.804	2.548	2.801	-0.203
OP2 (with selection)	3.158	2.912	2.634	2.862	-0.296

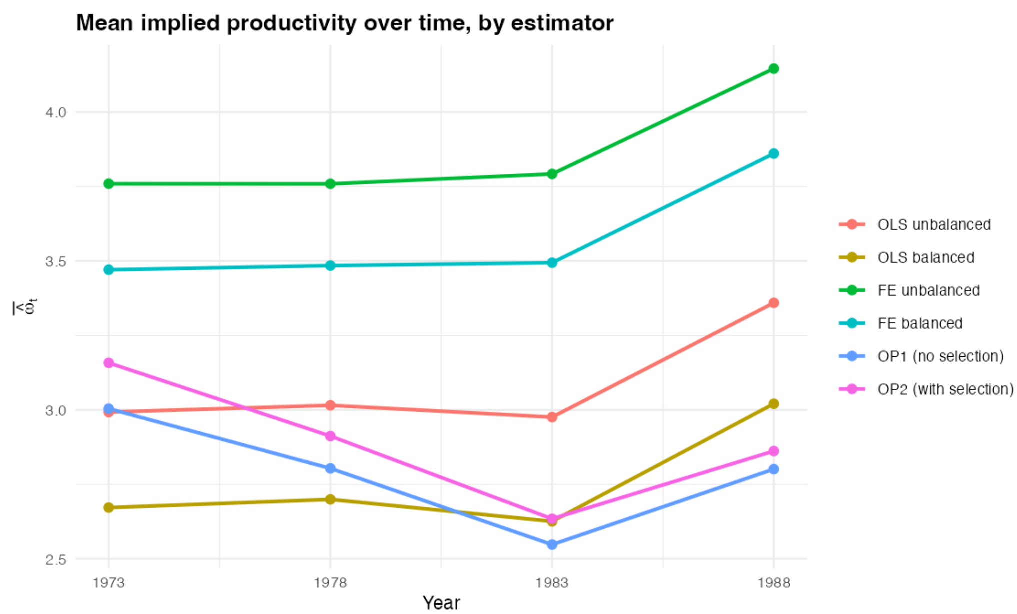

Figure 3: Mean implied productivity $\overline{\omega}_t^m$ by year, all six estimators. The OLS/FE specifications and OP1/OP2 give qualitatively different stories about the 1973–1988 productivity trend.

The cleanest reading of this divergence is selection. Through the 1973–1988 window the manufacturing panel experienced the Volcker disinflation and the mid-1980s industrial shakeout (Section 3: only 214 of 1,400 firms survived all four waves). Survivors had productivity draws systematically above the exit threshold, so a panel that pools survivors and exiters without correcting for selection mechanically shows "rising productivity" as the low-productivity firms drop out of later waves. OP corrects for this by conditioning on $\widehat{P}_{it}$, the survival probability fitted in Section 7. After conditioning, what looked like productivity growth is recovered as compositional change: the average level of productivity for a *fixed* firm did not rise; the average level for the *observed* firms rose only because the worst firms exited. The OP-implied path of *actual* productivity is mildly declining, consistent with the 1970s–80s U.S. manufacturing-TFP slowdown documented in the productivity-decade literature (e.g., Baily, 1986).

Figure 4 overlays the demeaned productivity densities of all six estimators on one axis. The OP shapes have visibly heavier left tails and broader support; the OLS/FE shapes are tighter, more bell-shaped, and miss the extreme low-productivity firms whose exit drives the selection story in the first place.

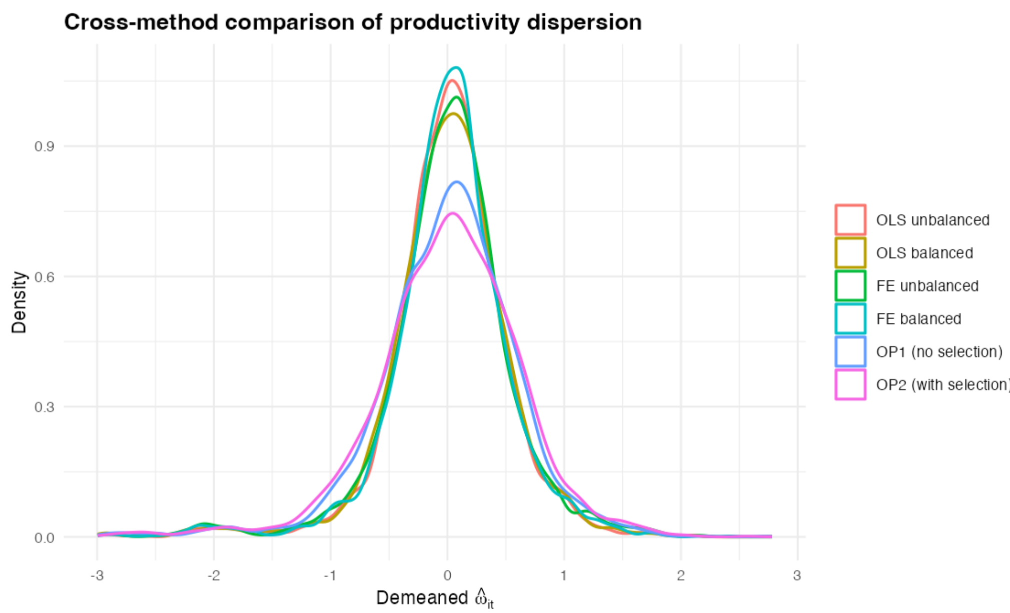

Figure 4: Demeaned productivity distributions, all six estimators. OP1 and OP2 (greens) sit visibly wider, with a heavier left tail capturing low-ω firms that the OLS/FE specifications wash out.

10 IMPLIED MARKUPS

Theory. Hall (1988) shows that a cost-minimizing firm with a flexible variable input X^V and an output elasticity $\theta_{X^V} = \partial \ln Q / \partial \ln X^V$ has a price-cost markup

$$\mu_{it} = \frac{P_{it}}{\text{MC}_{it}} = \frac{\theta_{X^V}}{s_{it}^V}, \qquad s_{it}^V \equiv \frac{W_{X^V}\, X_{it}^V}{P_{it}\, Q_{it}}, \tag{8}$$

where s^V is the variable input's share of revenue. De Loecker and Warzynski (2012) note that this requires only (i) cost minimization with respect to the flexible input and (ii) static optimality of X^V, so the markup can be backed out one observation at a time from the input-share inverse and the production-function elasticity. Taking labor as the flexible input and assuming Cobb–Douglas output ($\theta_\ell = \beta_\ell$) gives equation (7). For a value-added specification, $\hat{\mu}$ is a markup over short-run marginal cost *net of intermediate materials*; under the Klette and Griliches (1996) critique this systematically differs from a markup over gross marginal cost, and the gap can be sizable in materials-intensive industries. I proceed with the value-added specification because that is the production function estimated in Q6–Q8, and note where this matters in the discussion below.

Distribution. Table 11 reports level statistics for each estimator; Figure 5 shows the (log-scaled) histograms.

Table 11: Distribution of implied markups $\hat{\mu}_{it} = \hat{\beta}_\ell^m \exp(y_{it})/[W_t \exp(\ell_{it})]$ on the full unbalanced sample ($N = 2{,}971$). Levels depend on the Wage normalization (annual vs. hourly); within-method dispersion, time path, and cross-method comparison are unit-invariant.

Estimator	N	Mean	SD	P10	P25	Median	P75	P90
OLS unbalanced	2971	2.373	1.824	1.076	1.499	1.994	2.707	3.734
OLS balanced	2971	1.874	1.440	0.849	1.183	1.574	2.137	2.948
FE unbalanced	2971	3.370	2.590	1.528	2.128	2.831	3.844	5.302
FE balanced	2971	3.086	2.372	1.399	1.949	2.593	3.521	4.856
OP1 (no selection)	2971	2.325	1.787	1.054	1.468	1.953	2.652	3.658
OP2 (with selection)	2971	2.325	1.787	1.054	1.468	1.953	2.652	3.658

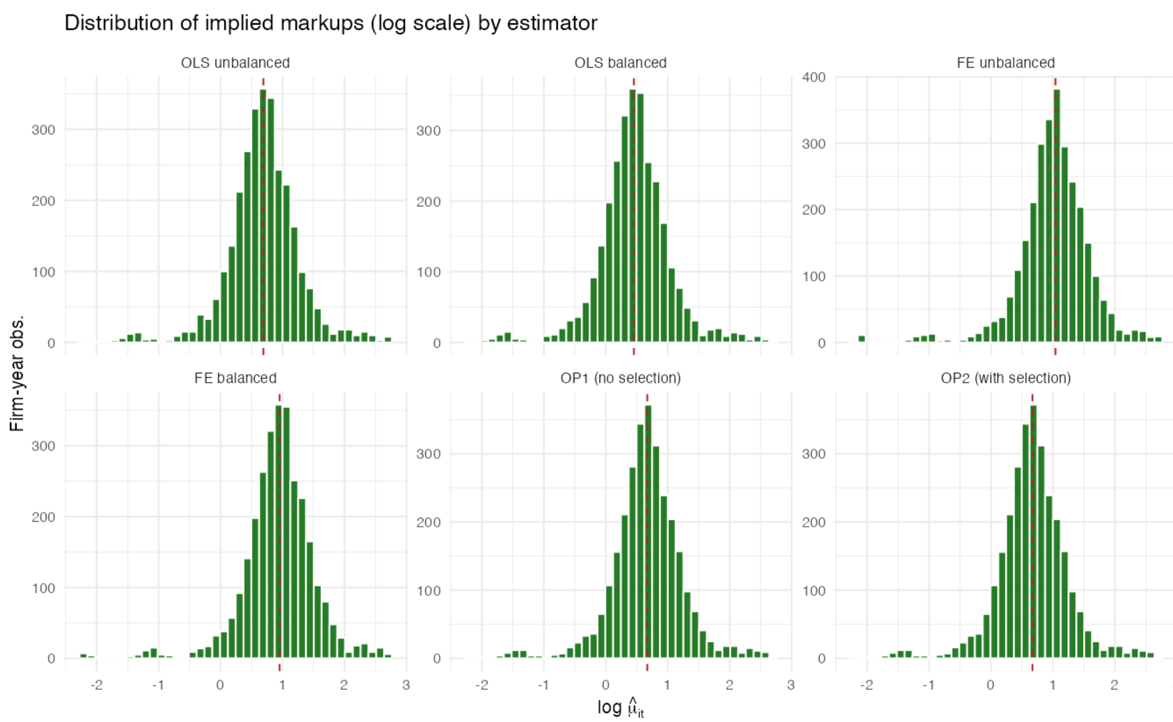

Figure 5: Six-panel histogram of implied markups (log scale; level statistics printed in-panel). Bin range trimmed at the 0.5/99.5 percentiles.

The most striking feature of the table is that **OP1 and OP2 are numerically identical** ($\overline{\hat{\mu}} = 2.33$, SD $= 1.79$, median $= 1.95$). This is by construction: equation (7) depends only on $\hat{\beta}_\ell$, and Section 8 recovered $\hat{\beta}_\ell^{\text{OP1}} = \hat{\beta}_\ell^{\text{OP2}} = 0.532$. The selection correction affects $\hat{\beta}_k$ ($0.360 \to 0.320$) but not $\hat{\beta}_\ell$, so it has no first-order effect on the markup. The substantive cross-method spread in markups therefore reduces to a spread across only four numbers ($\hat{\beta}_\ell \in \{0.429, 0.532, 0.707, 0.771\}$).

Median markups span median $\hat{\mu}^{\text{OLS-bal}} = 1.57$ to median $\hat{\mu}^{\text{FE-unbal}} = 2.83$, a factor of 1.80. Mechanically, $\hat{\mu} \propto \hat{\beta}_\ell$, so the FE specification's high labor coefficient (0.771) produces the highest markups and OLS-balanced's low labor coefficient (0.429) the lowest. **The cross-method spread in $\hat{\mu}$ inherits the cross-method spread in $\hat{\beta}_\ell$, and is therefore a quantitative restatement of the simultaneity-correction direction from Part 2 rather than independent information about market power.**

10.1 Outliers

The OP2 markup distribution has thick tails (Figure 5, bottom-right panel): $p_{10}/\text{median} \approx 0.54$ and $p_{90}/\text{median} \approx 1.87$. Table 12 lists the extreme tails. The smallest five markups are firm-years with $\hat{\mu} \in [0.062, 0.073]$, more than an order of magnitude below the median; the largest five sit at $\hat{\mu} \in [17.3, 20.4]$. **Both tails are concentrated in a small number of industries, and both are theoretically informative.** The bottom five are all in SIC 357 (Computer & Office Equipment) in 1973, firm-years where value-added per worker fell far below the wage, mechanically delivering $\hat{\mu} \ll 1$. This is impossible under the cost-minimization theory and reflects either measurement error in *ldsal* (small startup-era computer firms posting near-zero deflated value added with a non-trivial headcount), or a violation of static labor-input optimality (R&D-heavy firms hoarding skilled labor in advance of the PC takeoff). The top five are concentrated in SIC 291 (Petroleum Refining) and SIC 289 (Industrial Organic Chemicals) in 1973–1983, both notoriously capital-intensive, low-labor-share industries where revenue per worker is enormous relative to the wage bill, so $\exp(\textit{ldsal})/[W_t \exp(\textit{lemp})]$ blows up. Substantively, refining and bulk chemicals do command real price-cost wedges from sunk-capital scale economies, but $\hat{\mu} \approx 20$ is far above the textbook expectation, and the right interpretation is that the value-added Cobb–Douglas specification overstates the labor elasticity for these capital-dominated industries.

Table 12: Markup outliers under OP2 (with selection). Bottom 5 (smallest $\hat{\mu}$) and top 5 (largest $\hat{\mu}$) firm-years; firm IDs and industry codes shown.

Rank	Firm `index`	SIC3	Year	$\hat{\mu}$	$\log \hat{\mu}$
Bottom 5					
1	1279	357	1973	0.0621	-2.779
2	396	357	1973	0.0661	-2.717
3	1006	357	1973	0.0718	-2.633
4	905	357	1973	0.0725	-2.623
5	1349	357	1973	0.0728	-2.619
Top 5					
1	822	291	1973	17.3320	2.853
2	263	291	1973	17.8781	2.884
3	1274	291	1983	19.2181	2.956
4	774	289	1978	19.5926	2.975
5	263	291	1978	20.3819	3.015

A defensible cleanup would truncate at the 1st/99th percentiles and report robust statistics (median, IQR) for the headlines, which is already what the discussion above uses.

10.2 Heterogeneity by Industry

Table 13 and Figure 6 show medians for five SIC3 industries: 357 (Computer & Office Equipment, $n = 195$), 382 (Laboratory Instruments, $n = 189$), 367 (Electronic Components, $n = 159$), 283 (Drugs/Pharmaceuticals, $n = 137$), and 371 (Motor Vehicles, $n = 122$).

Table 13: Median implied markup $\hat{\mu}_{it}$ across five 3-digit SIC industries, by estimator. Industry names: 357 = Computer & Office Equipment; 382 = Laboratory Instruments; 367 = Electronic Components; 283 = Drugs; 371 = Motor Vehicles.

Estimator (n)	SIC 283 (137)	SIC 357 (195)	SIC 367 (159)	SIC 371 (122)	SIC 382 (189)
OLS unbalanced	2.269	0.834	1.049	2.140	1.654
OLS balanced	1.792	0.658	0.828	1.689	1.306
FE unbalanced	3.222	1.184	1.490	3.039	2.349
FE balanced	2.951	1.084	1.364	2.783	2.151
OP1 (no selection)	2.223	0.817	1.028	2.096	1.620
OP2 (with selection)	2.223	0.817	1.028	2.096	1.620

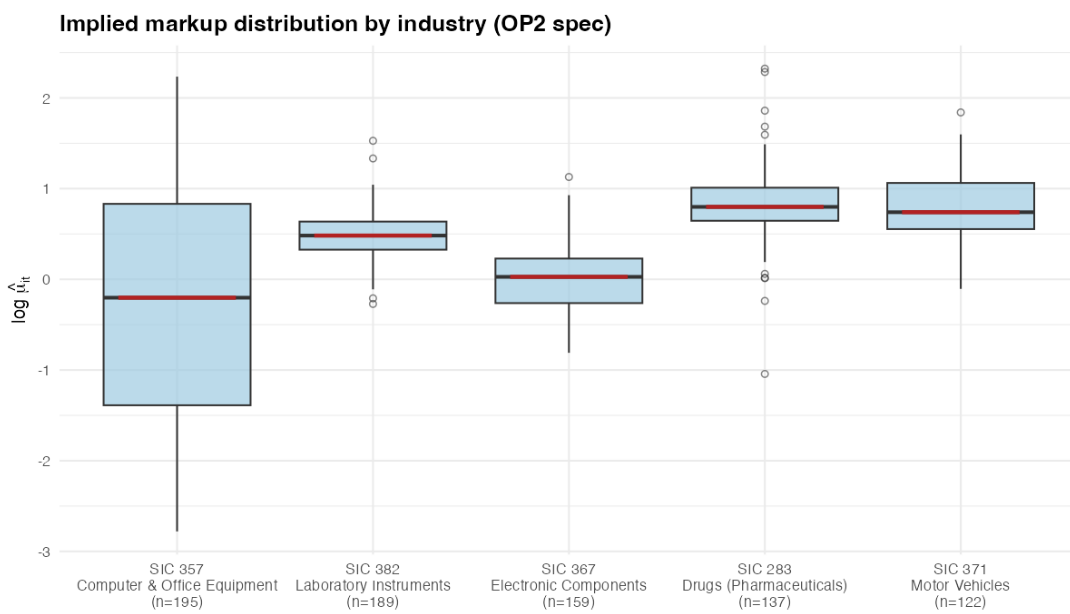

Figure 6: Distribution of log $\hat{\mu}_{it}$ across five industries (OP2 specification). Box = IQR; whiskers = 1.5 × IQR; circles = beyond-fence outliers. Median in firebrick.

Under OP2, the rank ordering is **drugs** $\succ$ **motor vehicles** $\succ$ **lab instruments** $\succ$ **electronic components** $\succ$ **computers** ($\hat{\mu} = 2.23$, 2.10, 1.63, 1.03, 0.82 respectively). Each of these is consistent with the industry's competitive structure in the 1973–1988 window:

- *Drugs (SIC 283)*, highest markup (2.22). Patent-protected blockbuster era; the major pharma houses (Merck, Pfizer, Lilly, Bristol-Myers) ran at marginal-cost-to-price ratios that the IO literature has consistently measured well above one for branded drugs (Scherer, 1993).
- *Motor Vehicles (SIC 371)*, second-highest (2.10). The 1973–1988 window is the Big 3 oligopoly under Japanese import competition; markups remained above one despite the entry threat,

consistent with the structure-conduct-performance literature on U.S. autos in this era.

- *Lab Instruments (SIC 382)*, mid-tier (1.62). Differentiated specialty equipment with switching costs but no patent moat.
- *Electronic Components (SIC 367)*, near unity (1.03). Commodity components (passives, connectors) faced intense international competition starting in the late 1970s.
- *Computer & Office Equipment (SIC 357)*, lowest (0.82). This industry sits below the theoretical floor of $\hat{\mu} = 1$. The window covers the mainframe-to-PC transition: IBM's dominant pricing in 1973–1978 (right tail in Figure 6) collapses into the price-war PC-clone era of 1983–1988 (long left tail). The wide spread in Figure 6, IQR more than twice that of any other industry, is itself the substantive finding: "computers" in this panel is two industries glued together.

10.3 Markups Over Time

Table 14 and Figure 7 show the year-by-year median for each estimator.

Table 14: Median implied markup $\hat{\mu}_t^m$ by year, all six estimators. Percent change is over the full 1973–1988 window.

Estimator	1973	1978	1983	1988	%Δ (88−73)
OLS unbalanced	2.301	1.918	1.807	2.051	-10.9%
OLS balanced	1.817	1.514	1.426	1.619	-10.9%
FE unbalanced	3.267	2.723	2.565	2.912	-10.9%
FE balanced	2.992	2.494	2.349	2.667	-10.9%
OP1 (no selection)	2.254	1.879	1.770	2.009	-10.9%
OP2 (with selection)	2.254	1.879	1.770	2.009	-10.9%

All six estimators give the same time pattern, a U-shape with the trough in 1983, because all six share the same data on *ldsal*, *lemp*, and W_t, and the time-varying component of $\hat{\mu}_{it}$ comes overwhelmingly from these inputs. Specifically, the wage series rises monotonically over the panel ($W_{73} = 18.4$, $W_{78} = 21.9$, $W_{83} = 23.6$, $W_{88} = 28.9$), and equation (7) puts W_t in the denominator. Median value-added per worker did not keep pace until 1988 (when productivity rebounded, per Figure 3 in the OLS/FE readings). The total 1973 $\to$ 1988 change is -10.9% across every estimator, with the trough in 1983 reflecting the depth of the post-Volcker recession.

This pattern is qualitatively the *opposite* of De Loecker, Eeckhout, and Unger (2020)'s headline finding that U.S. aggregate markups rose from ≈ 1.21 in 1980 to ≈ 1.61 in 2014. The reconciliation is straightforward: the De Loecker, Eeckhout, and Unger rise is concentrated in the post-1990 window, and the 1973–1988 panel here ends before that rise began. The U-shape we recover is consistent with the markup decline of the late 1970s and early 1980s that even the De Loecker, Eeckhout, and Unger series shows on the way down.

10.4 Cross-Method Comparison

Within each year, the cross-method spread in median $\hat{\mu}$ is roughly a factor of ≈ 1.8 (FE-unbal vs. OLS-bal), but the time pattern is identical across methods, and the within-method dispersion (e.g., OP2 $p_{10}/p_{90} = 0.54/3.67 \approx 6.8\times$) is far larger than the between-method dispersion. **The**

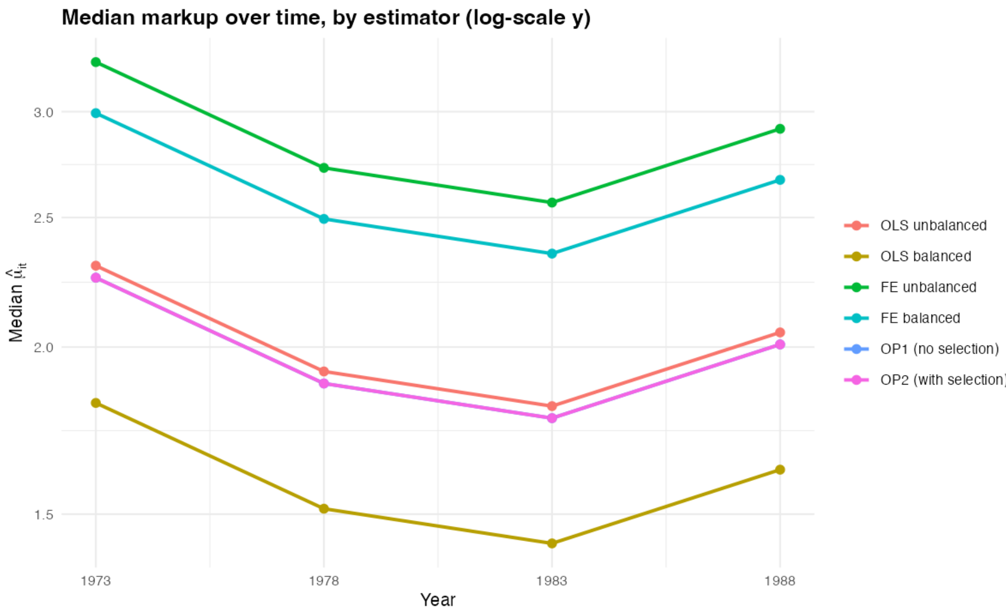

Figure 7: Median implied markup by year, all six estimators (log-scale y to compress the level spread). All six estimators trace the same U-shape: median markup falls 1973 $\rightarrow$ 1983 and recovers partially 1983 $\rightarrow$ 1988.

choice of production-function estimator changes the level of the markup distribution; it does not change the time path, the industry ranking, or the dispersion. For a policy claim about which industries are highly marked-up, when markups peaked, or whether the dispersion of market power is wide, the answer is robust across estimators. For a policy claim about the *absolute* level of markups, the answer is sensitive to which production-function specification is in hand, and that sensitivity is dominated by the labor coefficient $\hat{\beta}_\ell$, which Part 2 showed is itself sensitive to the Marschak–Andrews simultaneity correction (modest at five-year frequency here, but in principle large).

CONCLUSION

The three parts of this problem set fit together as a single argument about what selection bias costs you when you skip it. Part 1 documents that the panel is heavily unbalanced and that survivors differ from exiters along every input dimension by roughly a factor of four, establishing the selection problem as a data fact rather than an econometric abstraction. Part 2 estimates the production function six different ways and shows that the labor coefficient is largely robust across methods ($\hat{\beta}_\ell \in [0.43, 0.77]$ with the OP estimate at 0.532) while the capital coefficient is sensitive ($\hat{\beta}_k \in [0.19, 0.53]$, with both OP estimates *below* the OLS unbalanced baseline, the opposite direction from Olley and Pakes (1996)'s telecoms-equipment finding, driven by this dataset's multi-industry pooling). Part 3 uses the estimates to recover firm productivity and markups, finding that the Olley–Pakes selection correction matters quantitatively for productivity (the time-trend reverses sign relative to naive estimators; cross-firm dispersion widens by $\approx 15\%$) but not for markups (OP1 and OP2 are numerically identical because $\hat{\mu} \propto \hat{\beta}_\ell$ and the selection correction doesn't move $\hat{\beta}_\ell$). Markups display the expected industry ordering, drugs $>$ autos $>$

lab instruments $>$ electronics $>$ computers, with computers anomalously below unity because the panel pools the mainframe-incumbency and PC-clone-price-war eras. The policy-analysis takeaway is that estimator choice changes the *level* of inferred productivity and market power but not its industry ranking, time path, or dispersion structure. The cleanest test of whether the OP correction is worth the trouble on this kind of panel is whether it changes a conclusion you would otherwise draw from OLS. Here, productivity growth flips sign; markups, industry rankings, and the U-shape over time do not. That is the boundary of the selection correction's substantive contribution in this dataset, and it is a useful boundary to know before applying OP to a different panel where exit may bite differently.

A REPRODUCIBILITY

The full pipeline is reproducible from `prodfncData.dta` alone. Open R in the assignment directory and source `Dalis_Abdallah_main.R`. Each script writes its outputs to `./figures/` (PNG, 150 dpi) and `./tables/` (LaTeX). Required packages: `haven`, `dplyr`, `tidyr`, `ggplot2`, `xtable`, `fixest`, `modelsummary`, `kableExtra`, `tibble`. Bootstrap seed is 514; bootstrap reps $B = 200$.

Table 15: R pipeline. Sourcing the master script runs all six in order; each downstream script will re-source its predecessors only if the required objects are missing from the workspace.

Script	Maps to	Produces
`01_descriptives.R`	Q1–Q5	`figures/hist_topind_ldsal.png`; `tables/industry_top10.tex`, `balance_panel.tex`, `sumstats_full.tex`, `sumstats_balanced.tex`; objects `df`, `df_bal`.
`02_ols_fe.R`	Q6	`tables/ols_fe_coefs.tex`; objects `ols_unb`, `ols_bal`, `fe_unb`, `fe_bal`.
`03_probit_survival.R`	Q7	`tables/probit_survival.tex`; objects `prob_simple`, `prob_poly`, `df_age` (panel augmented with `age`, `survival`, $\widehat{P}_{it}$).
`04_olley_pakes.R`	Q8	`tables/op_estimates.tex`; object `op_coefs` (OP1 and OP2 point estimates, bootstrap draws, inference).
`05_productivity.R`	Q9	`figures/prod_hist_by_estimator.png`, `prod_mean_by_year.png`, `prod_density_overlay.png`; `tables/prod_summary.tex`, `prod_by_year.tex`; object `prod_panel`.
`06_markups.R`	Q10	`figures/markup_hist.png`, `markup_by_year.png`, `markup_by_sic.png`; `tables/markup_summary.tex`, `markup_outliers.tex`, `markup_by_sic.tex`, `markup_by_year.tex`.

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